



考虑一般的带约束条件的优化问题：

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & h_i(x) \leq 0, \quad i = 1, \dots, m \\ & \ell_j(x) = 0, \quad j = 1, \dots, r \end{aligned}$$

拉格朗日乘子主要用于解决带约束条件下的优化问题。其基本思想是通过引入拉格朗日乘子，将含有 n 个变量、 $(m+r)$ 个约束条件的约束优化问题转化为含有 $(n+m+r)$ 个变量的无约束优化问题。

$$L(x, u, v) = f(x) + \sum_{i=1}^m u_i h_i(x) + \sum_{j=1}^r v_j \ell_j(x)$$

New variables $u \in \mathbb{R}^m, v \in \mathbb{R}^r$, with $u \geq 0$



Important property: for any $u \geq 0$ and v ,

$$f(x) \geq L(x, u, v) \quad \text{at each feasible } x$$

Why? For feasible x ,

$$L(x, u, v) = f(x) + \sum_{i=1}^m u_i \underbrace{h_i(x)}_{\leq 0} + \sum_{j=1}^r v_j \underbrace{\ell_j(x)}_{=0} \leq f(x)$$



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Let C denote primal feasible set, f^* denote primal optimal value.
Minimizing $L(x, u, v)$ over all $x \in \mathbb{R}^n$ gives a lower bound:

$$f^* \geq \min_{x \in C} L(x, u, v) \geq \min_{x \in \mathbb{R}^n} L(x, u, v) := g(u, v)$$

We call $g(u, v)$ the **Lagrange dual function**, and it gives a lower bound on f^* for any $u \geq 0$ and v , called dual feasible u, v



Example:

Consider quadratic program (QP, step up from LP!)

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \frac{1}{2} x^T Q x + c^T x \\ \text{subject to} \quad & Ax = b, x \geq 0 \end{aligned}$$

where $Q \succ 0$. Lagrangian:

$$L(x, u, v) = \frac{1}{2} x^T Q x + c^T x - u^T x + v^T (Ax - b)$$

Lagrange dual function:

$$g(u, v) = \min_{x \in \mathbb{R}^n} L(x, u, v) = -\frac{1}{2} (c - u + A^T v)^T Q^{-1} (c - u + A^T v) - b^T v$$

For any $u \geq 0$ and any v , this is lower a bound on primal optimal value f^*



Given primal problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & h_i(x) \leq 0, \quad i = 1, \dots, m \\ & \ell_j(x) = 0, \quad j = 1, \dots, r \end{aligned}$$

Our constructed dual function $g(u, v)$ satisfies $f^* \geq g(u, v)$ for all $u \geq 0$ and v . Hence best lower bound is given by maximizing $g(u, v)$ over all dual feasible u, v , yielding **Lagrange dual problem**:

$$\begin{aligned} \max_{u \in \mathbb{R}^m, v \in \mathbb{R}^r} \quad & g(u, v) \\ \text{subject to} \quad & u \geq 0 \end{aligned}$$

Key property, called **weak duality**: if dual optimal value g^* , then

$$f^* \geq g^*$$

Note that this always holds (even if primal problem is nonconvex)



Another key property: the dual problem is a **convex optimization** problem (as written, it is a concave maximization problem)

Again, this is always true (even when primal problem is not convex)

By definition:

$$\begin{aligned} g(u, v) &= \min_{x \in \mathbb{R}^n} \left\{ f(x) + \sum_{i=1}^m u_i h_i(x) + \sum_{j=1}^r v_j \ell_j(x) \right\} \\ &= - \max_{x \in \mathbb{R}^n} \underbrace{\left\{ -f(x) - \sum_{i=1}^m u_i h_i(x) - \sum_{j=1}^r v_j \ell_j(x) \right\}}_{\text{pointwise maximum of convex functions in } (u, v)} \end{aligned}$$

I.e., g is concave in (u, v) , and $u \geq 0$ is a convex constraint, hence dual problem is a concave maximization problem



Recall that we always have $f^* \geq g^*$ (weak duality). On the other hand, in some problems we have observed that actually

$$f^* = g^*$$

which is called **strong duality**

Conjunction: 如果 x^* 和 u^*, v^* 分别是原问题和对偶问题的可行解, 并且 $f(x^*) = g(u^*, v^*)$, 则 x^* 和 u^*, v^* 分别是原问题和对偶问题的最优解。



Slater's condition: if the primal is a convex problem (i.e., f and $h_1, \dots, h_m$ are convex, $\ell_1, \dots, \ell_r$ are affine), and there exists at least one strictly feasible $x \in \mathbb{R}^n$, meaning

$$h_1(x) < 0, \dots, h_m(x) < 0 \quad \text{and} \quad \ell_1(x) = 0, \dots, \ell_r(x) = 0$$

then strong duality holds

This is a pretty weak condition. (And it can be further refined: need strict inequalities only over functions h_i that are not affine)

Given general problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & h_i(x) \leq 0, \quad i = 1, \dots, m \\ & \ell_j(x) = 0, \quad j = 1, \dots, r \end{aligned}$$

The **Karush-Kuhn-Tucker conditions** or **KKT conditions** are:

- $0 \in \partial f(x) + \sum_{i=1}^m u_i \partial h_i(x) + \sum_{j=1}^r v_j \partial \ell_j(x)$ (stationarity)
- $u_i \cdot h_i(x) = 0$ for all i (complementary slackness)
- $h_i(x) \leq 0, \ell_j(x) = 0$ for all i, j (primal feasibility)
- $u_i \geq 0$ for all i (dual feasibility)

Necessity

Let x^* and u^*, v^* be primal and dual solutions with zero duality gap (strong duality holds, e.g., under Slater's condition). Then

$$\begin{aligned} f(x^*) &= g(u^*, v^*) \\ &= \min_{x \in \mathbb{R}^n} f(x) + \sum_{i=1}^m u_i^* h_i(x) + \sum_{j=1}^r v_j^* \ell_j(x) \\ &\leq f(x^*) + \sum_{i=1}^m u_i^* h_i(x^*) + \sum_{j=1}^r v_j^* \ell_j(x^*) \\ &\leq f(x^*) \end{aligned}$$

In other words, all these inequalities are actually equalities



Two things to learn from this:

- The point x^* minimizes $L(x, u^*, v^*)$ over $x \in \mathbb{R}^n$. Hence the subdifferential of $L(x, u^*, v^*)$ must contain 0 at $x = x^*$ —this is exactly the **stationarity** condition
- We must have $\sum_{i=1}^m u_i^* h_i(x^*) = 0$, and since each term here is ≤ 0 , this implies $u_i^* h_i(x^*) = 0$ for every i —this is exactly **complementary slackness**

Primal and dual feasibility obviously hold. Hence, we've shown:

If x^* and u^*, v^* are primal and dual solutions, with zero duality gap, then x^*, u^*, v^* satisfy the KKT conditions

(Note that this statement assumes nothing a priori about convexity of our problem, i.e. of f, h_i, ℓ_j)



Sufficiency

If there exists x^*, u^*, v^* that satisfy the KKT conditions, then

$$\begin{aligned} g(u^*, v^*) &= f(x^*) + \sum_{i=1}^m u_i^* h_i(x^*) + \sum_{j=1}^r v_j^* \ell_j(x^*) \\ &= f(x^*) \end{aligned}$$

where the first equality holds from stationarity, and the second holds from complementary slackness



Therefore duality gap is zero (and x^* and u^*, v^* are primal and dual feasible) so x^* and u^*, v^* are primal and dual optimal. I.e., we've shown:

When the primal problem is convex

If x^* and u^*, v^* satisfy the KKT conditions, then x^* and u^*, v^* are primal and dual solutions



Therefore

In summary, KKT conditions:

- always sufficient When the primal problem is convex
- necessary under strong duality

For a problem with strong duality (e.g., assume Slater's condition: convex problem and there exists x strictly satisfying non-affine inequality constraints),

x^* and u^*, v^* are primal and dual solutions

$\Leftrightarrow x^*$ and u^*, v^* satisfy the KKT conditions

Solving the primal via the dual

An important consequence of stationarity: under strong duality, given a dual solution u^*, v^* , any primal solution x^* solves

$$\min_{x \in \mathbb{R}^n} f(x) + \sum_{i=1}^m u_i^* h_i(x) + \sum_{j=1}^r v_j^* \ell_j(x)$$

Often, solutions of this unconstrained problem can be expressed explicitly, giving an explicit **characterization** of primal solutions (from dual solutions)

Furthermore, suppose the solution of this problem is unique; then it must be the primal solution x^*

This can be very helpful when the dual is easier to solve than the primal